CMS eCQM CMS334v7 (PC-02), CBE ID 0471e, 2026 REPORTING PERIOD

Cesarean Birth

The published eCQM specification, implemented in SQL and tested on 15 synthetic deliveries

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What the measure is. CMS334 is the eCQM version of PC-02: the share of nulliparous patients with a term, singleton baby in a vertex position (NTSV) who deliver by cesarean birth. Steward: The Joint Commission. Proportion outcome measure, no stratification or risk adjustment. A decreased score indicates improvement; the specification adds that the steward does not encourage inappropriately low rates. **Every patient, hospital, date and record on this page is synthetic. No CMS, OBI, hospital or patient data was used.**

Source of record. CMS334v7, eCQM version 7.1.000, measurement period January 1 to December 31, 2026, on the eCQI Resource Center: ecqi.healthit.gov/ecqm/hosp-inpt/2026/cms0334v7, file CMS334-v7.1.000-QDM.html. Citations name sections of that file: **PC** Population Criteria, **Def** Definitions, **Fn** Functions, **G** Guidance. CMS334v8 (8.1.000) is posted for the 2027 reporting period; the logic cited here reads the same in it.

MEASURE LOGIC AS IMPLEMENTED

1. **Initial population.** An "Encounter Inpatient" stay that ends in the measurement period, for a patient 8 or older and under 65 on the admission date, with a "Delivery Procedures" procedure starting during the stay. [PC Initial Population; Def Global.Inpatient Encounter, PCMaternal.Encounter With Age Range, PCMaternal.Delivery Encounter With Age Range]
2. **Term.** Gestational age of 37 weeks or more, from the first source available: (a) calculated, (280 - days from delivery date to estimated due date) div 7; (b) the last "Estimated Gestational Age at Delivery" result from 24 hours before delivery to the end of the delivery day; (c) an encounter diagnosis in "37 to 42 Plus Weeks Gestation". [Def Delivery Encounter With Gestational Age Greater Than Or Equal To 37 Weeks; Fn PCMaternal.CalculatedGestationalAge, PCMaternal.LastEstimatedGestationalAge; G]
3. **Singleton.** An encounter diagnosis in "Delivery of Singleton". [Def Encounter With Singleton Delivery]
4. **Nulliparous.** Gravida = 1, or parity = 0, or preterm births = 0 and term births = 0, each the last value charted in the 42 weeks before the time of delivery. Steps 2 to 4 give the denominator. [PC Denominator; Fn LastGravida, LastParity, LastHistoryPretermBirth, LastHistoryTermBirth; header Definition]
5. **Denominator exclusions.** The last "Abnormal Presentation" assessment at or before delivery falls during the stay, or an encounter diagnosis is in "Abnormal Presentation", "Placenta Accreta Spectrum Previa or Vasa Previa" or "Genital Herpes". Vertex is not tested directly: the Guidance says it is modeled implicitly through this exclusion. [PC Denominator Exclusions; Def Encounter With Abnormal Presentation; G]
6. **Numerator.** A "Cesarean Birth" procedure during the stay. Excluded stays are removed before the rate is taken. [PC Numerator; Def Delivery Encounter With Cesarean Birth]

Value sets read, by name and OID. Encounter Inpatient 2.16.840.1.113883.3.666.5.307; Delivery Procedures 2.16.840.1.113762.1.4.1045.59; Estimated Gestational Age at Delivery 2.16.840.1.113762.1.4.1045.26; 37 to 42 Plus Weeks Gestation 2.16.840.1.113762.1.4.1110.68; Delivery of Singleton 2.16.840.1.113762.1.4.1045.99; Abnormal Presentation 2.16.840.1.113762.1.4.1045.105; Placenta Accreta Spectrum Previa or Vasa Previa 2.16.840.1.113762.1.4.1110.37; Genital Herpes 2.16.840.1.113883.3.464.1003.110.12.1049; Cesarean Birth 2.16.840.1.113883.3.117.1.7.1.282. The other elements use direct reference codes.

TEST CASES: 15 SYNTHETIC DELIVERIES AT 2 SYNTHETIC HOSPITALS

GA gestational age, G gravida, P parity, CS cesarean birth, VB vaginal birth

ID	Hosp	Rule tested	Synthetic delivery	Birth	IP	Den	Excl	Num
SYN-D01	A	Baseline	G1 P0, calculated 39 weeks, singleton	VB	I	D	.	.
SYN-D02	A	Term boundary	Due date 21 days after delivery: (280 - 21) div 7 = 37. Gravida 1, parity not recorded	CS	I	D	.	N
SYN-D03	A	Term boundary	Due date 22 days after delivery: (280 - 22) div 7 = 36. Discrete field says 37, not read	VB	I	.	.	.
SYN-D04	A	GA source	No due date. Discrete gestational age field says 38 weeks, 2 hours before delivery	VB	I	D	.	.
SYN-D05	B	GA source	No due date or discrete field. Encounter diagnosis in "37 to 42 Plus Weeks Gestation"	CS	I	D	.	N
SYN-D06	A	Multiples	Multiple birth, calculated 37 weeks: no "Delivery of Singleton" diagnosis	CS	I	.	.	.
SYN-D07	B	Parity	G2 P0: parity 0 qualifies with gravida 2	VB	I	D	.	.
SYN-D08	B	Parity	G2 P1 with one prior term birth: not nulliparous	CS	I	.	.	.
SYN-D09	B	Parity	No gravida or parity. Preterm 0 and term 0 before delivery; term set to 1 after delivery	CS	I	D	.	N
SYN-D10	A	Presentation	"Abnormal Presentation" assessment during the stay, before delivery	CS	I	D	E	.
SYN-D11	B	Presentation	"Abnormal Presentation" as an encounter diagnosis, no assessment	CS	I	D	E	.
SYN-D12	B	Presentation	"Abnormal Presentation" assessed only at a prenatal visit 20 days before admission	VB	I	D	.	.
SYN-D13	A	Exclusion	Encounter diagnosis in "Placenta Accreta Spectrum Previa or Vasa Previa"	CS	I	D	E	.
SYN-D14	B	Exclusion	Encounter diagnosis in "Genital Herpes"	CS	I	D	E	.
SYN-D15	A	Period	Delivered Dec 31, 2026; discharged Jan 2, 2027, after the measurement period	VB	.	.	.	.

Key. **I** initial population, **D** denominator, **E** denominator exclusion, **N** numerator counted after exclusions, a dot = not in that population. Hosp A, B = SYN-HOSP-A, B.

Result: 15 of 15 cases pass. The expected result for each case was written out from the specification first, then the SQL was run. Separately, 16 single-rule edits to the SQL were each caught by exactly the cases named in advance. Net den is Den less Excl. On 15 synthetic deliveries the rates only check the arithmetic.

Synthetic hospital roll-up	IP	Den	Excl	Net den	Num	Num / Net den
SYN-HOSP-A	7	5	2	3	1	1 of 3 (33.3%)
SYN-HOSP-B	7	6	2	4	2	2 of 4 (50.0%)
Both	**14**	**11**	**4**	**7**	**3**	**3 of 7 (42.9%)**

Implementation notes. (1) "Days from delivery date to due date" follows the CQL "difference in days", which compares calendar dates and ignores the time of day: 22 days gives 36 weeks, 21 days gives 37 (SYN-D02, D03). (2) The gestational age sources are ordered. When a calculated value exists, the discrete field and the diagnosis codes are not read, including when the calculated value is under 37 (D03) [G]. (3) Gravida, parity and birth counts are read only before the time of delivery; the Guidance notes parity and birth counts may be updated during the delivery stay (D09) [G]. (4) The numerator definition is drawn from the denominator, so an excluded cesarean also meets it; the code removes exclusions before counting (D10, D11, D13, D14). (5) Stillbirth and multiple gestation are not separate logic steps; the Guidance says the live singleton requirement already rules them out [G].

What this does not cover. (1) The extension of the stay back through an emergency department or OB triage visit and an observation stay [Fn PCMaternal.HospitalizationWithEDOBTriageObservation]; here the stay runs from inpatient admission to discharge. (2) Value set contents: each synthetic record carries a value set name, and no codes were read. (3) Supplemental data elements, QRDA reporting, and elements recorded as a period. (4) Cases on the failing side of the age bounds, the delivery procedure requirement, and the edges of the 24-hour and 42-week windows. (5) The chart-abstracted PC-02 and OBI's own data definitions; this page is the CMS eCQM only.

About me. Pharmacist by training (B.Pharm, India; not US-licensed). Master of Health Informatics, University of Michigan (April 2026). I package CMS digital quality measures as executable knowledge objects with Dr. Allen Flynn (U-M Knowledge Systems Lab), starting with CMS108 in CQL, and have written SAS cohort and analysis code for an NIH-funded R01 at Michigan Medicine.

Code. github.com/gauravcb99/cms334-cesarean-birth. One SQL statement run in SQLite from Python, standard library only.